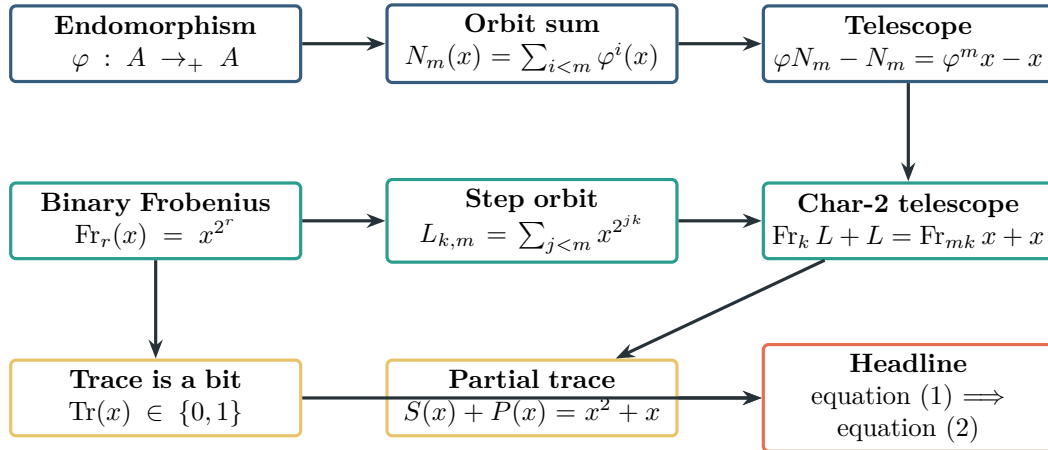


Dobbertin Theorem 1: Standalone LEGO Path

From an additive endomorphism to equation (1) $\Rightarrow$ equation (2)



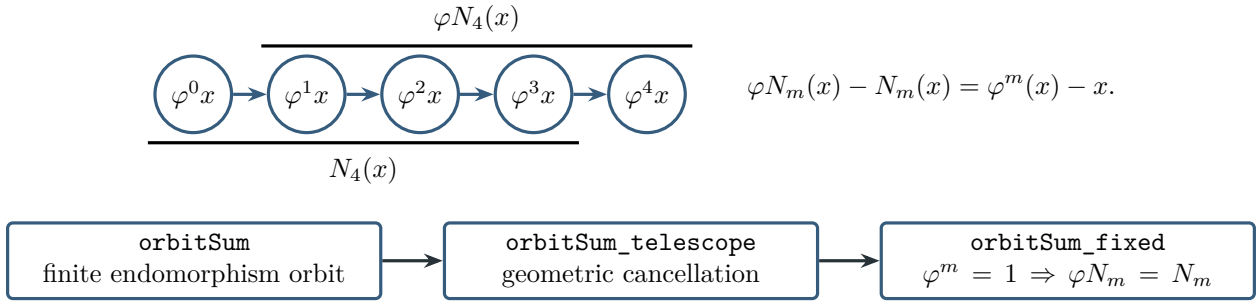
Scope. This minimal library proves the first linearization step used in Dobbertin's proof of Theorem 1. It deliberately stops before the later fibre classification and boundary-parity argument. The root module imports only Mathlib and the three modules in `DobbertinTheorem1Standalone/`.

1. The universal block: an orbit telescope

Let A be an additive commutative group and let $\varphi \in \text{End}(A)$. The only abstract construction is

$$N_m(x) = x + \varphi(x) + \cdots + \varphi^{m-1}(x).$$

Applying φ shifts every term one place. Subtraction cancels the interior:



The first arrow uses only the additive-group laws. The second is a one-line consequence when the orbit closes. No field, characteristic, or finiteness is present yet.

2. Specialization to binary Frobenius

Over $F = \mathbb{F}_{2^n}$, set $\text{Fr}_r(x) = x^{2^r}$. Characteristic two supplies additivity, while finite-field Fermat supplies the closed orbit:

$$\text{Fr}_r(x + y) = \text{Fr}_r(x) + \text{Fr}_r(y), \quad \text{Fr}_a(\text{Fr}_b(x)) = \text{Fr}_{a+b}(x), \quad \text{Fr}_n(x) = x.$$

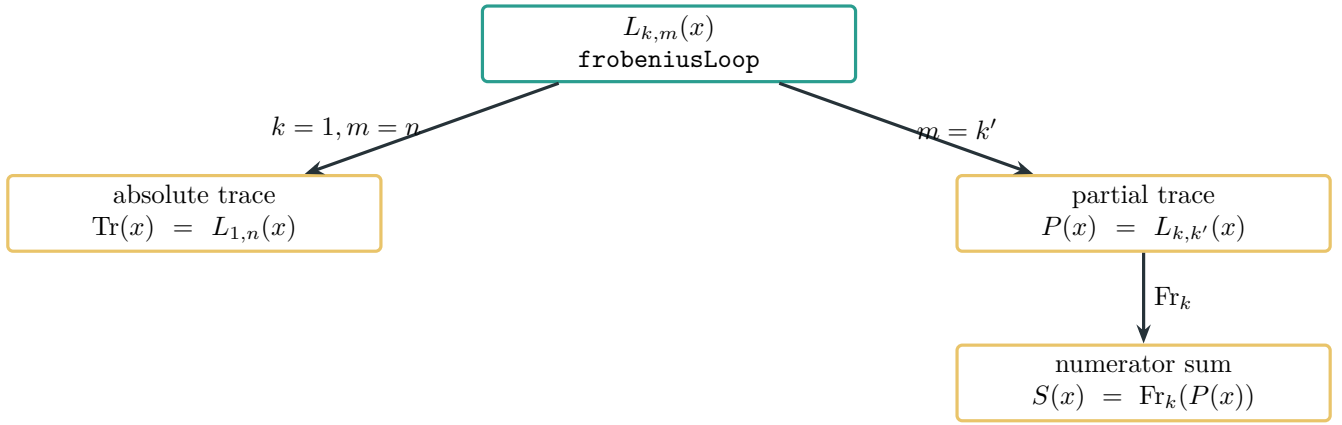
Thus Fr_k is an additive endomorphism and the abstract norm becomes

$$L_{k,m}(x) = \sum_{j=0}^{m-1} \text{Fr}_{jk}(x) = \sum_{j=0}^{m-1} x^{2^{jk}}.$$

In characteristic two, minus equals plus, so the universal telescope reads

$$\boxed{\text{Fr}_k(L_{k,m}(x)) + L_{k,m}(x) = \text{Fr}_{mk}(x) + x}.$$

3. The paper objects are settings of one loop

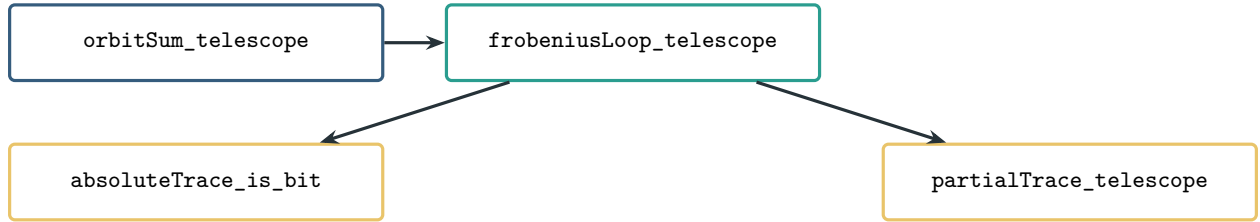


The full orbit is fixed by Frobenius. Therefore $\text{Tr}(x)^2 = \text{Tr}(x)$, and field factorization gives

$$\text{Tr}(x)(\text{Tr}(x) - 1) = 0 \implies \boxed{\text{Tr}(x) = 0 \text{ or } \text{Tr}(x) = 1}.$$

Assume $kk' \equiv 1 \pmod{n}$. Frobenius periodicity changes the far endpoint of the step-orbit telescope into $\text{Fr}_1(x) = x^2$:

$$\begin{aligned} S(x) + P(x) &= \text{Fr}_k(L_{k,k'}(x)) + L_{k,k'}(x) \\ &= \text{Fr}_{k'k}(x) + x \\ &= x^2 + x. \end{aligned}$$

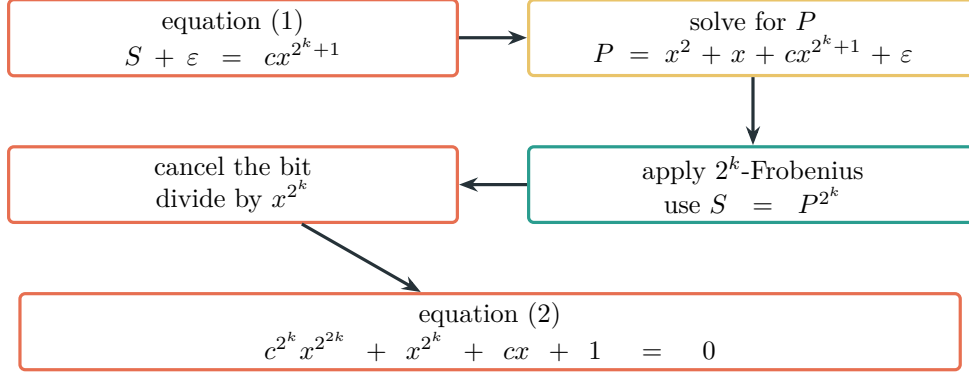


4. Equation (1) to equation (2)

For $\alpha \in \{0, 1\}$, the cleared fibre equation is represented as

$$cx^{2^k+1} = S(x) + \alpha \operatorname{Tr}(x). \quad (1)$$

Put $\varepsilon = \alpha \operatorname{Tr}(x)$. The trace result says $\varepsilon \in \{0, 1\}$, hence $\varepsilon^{2^k} = \varepsilon$. Combine (1) with $S + P = x^2 + x$ and $S = P^{2^k}$:



The division is the sole reason for the hypothesis $x \neq 0$. The final theorem is:

$$|F| = 2^n, \quad kk' \bmod n = 1, \quad \alpha \in \{0, 1\}, \quad x \neq 0, \\ \text{EquationOne}(n, k, k', \alpha, c, x) \implies \text{EquationTwoExpression}(k, c, x) = 0.$$

Lean headline:

`Dobbertin.Theorem1Standalone.`

`equationOne_implies_equationTwo.`

Files: `Endomorphism.lean` $\rightarrow$ `Frobenius.lean` $\rightarrow$ `Linearization.lean`.

Root: `DobbertinTheorem1Standalone.lean`.